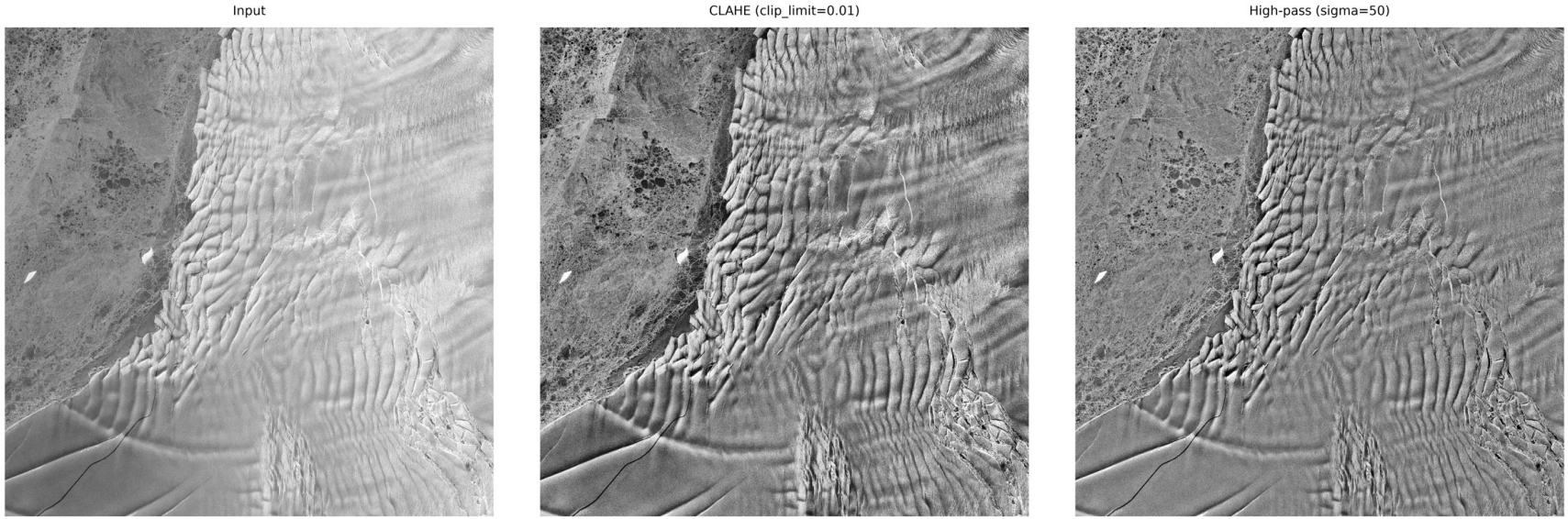
Data Augmentation Methods

CLAHE and high-pass filtering on NISAR tiles

Data Science Clinic
Alexandra Marcos



Frames to Tiles

~30,000 x 30,000

pixels in one NISAR frame (HHHH backscatter, float32)

4096 x 4096

pixels in one tile, identified by frame + row + column

0 = nodata

gray levels 1-255 are real data, 0 is outside the swath

1st-99th pct

per-frame stretch of the dB values onto gray levels 1-255

The same backscatter looks the same in every tile of a frame, because the stretch is computed once per frame, not per tile. Tiles are cut with the Label Studio tile server code, so they match what we label.

Import Image Functions

1

`load_tileserver`

scripts/enhance_contrast.py

Imports labelstudio/tileserver/tileserver.py without changing it. It sets the image and cache folders first, because the file reads them when imported.

2

`get_stats`

tileserver.py

One streaming pass over the .h5 file: the 1st and 99th percentile in dB, and the valid-pixel count of every 512-pixel block. Cached as a small JSON, so only the first run is slow.

3

`list_tiles`

tileserver.py

Lists the tiles with at least 5% valid pixels. It sums the cached block counts, so it never re-reads the image.

4

`render_tile`

tileserver.py

Reads only that tile's window, converts to dB, rescales to gray 1-255, and sets nodata to 0. Returns a uint8 array.

In the notebook: load_tiles() wraps get_stats + render_tile. choose_frames and choose_tiles (also in the script) add the seeded random pick.

Methods:

CLAHE

`clip_limit`

Stretches contrast in small local windows, so faint features in a dull area get brighter too.

Higher clip_limit = more contrast, but also more speckle noise.

`utils.augmentation.clahe (skimage equalize_adapthist)`

High-pass

`sigma`

Blurs the tile, subtracts the blur, and stretches the rest. Slow brightness changes go, small features like crevasses stay.

A small sigma can erase thin cracks. A large sigma keeps more large-scale brightness.

`utils.augmentation.high_pass (scipy gaussian_filter)`

Process to set clip_limit and sigma

1 Import libraries

Plus the script's functions (enhance_contrast).

2 Define a tile

A frame + row + column. load_tiles() returns the tiles.

3 Sample tiles

Random frame and tiles. SEED, N_FRAMES, N_TILES.

4 CLAHE sweep

clip_limit 0.005, 0.01, 0.02, 0.05. One figure per tile.

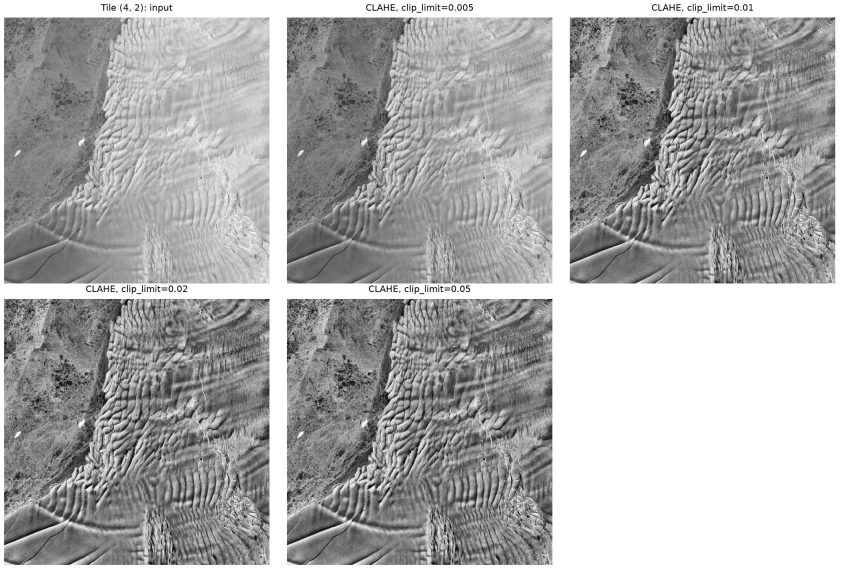
5 High-pass sweep

sigma 10, 25, 50, 100. One figure per tile.

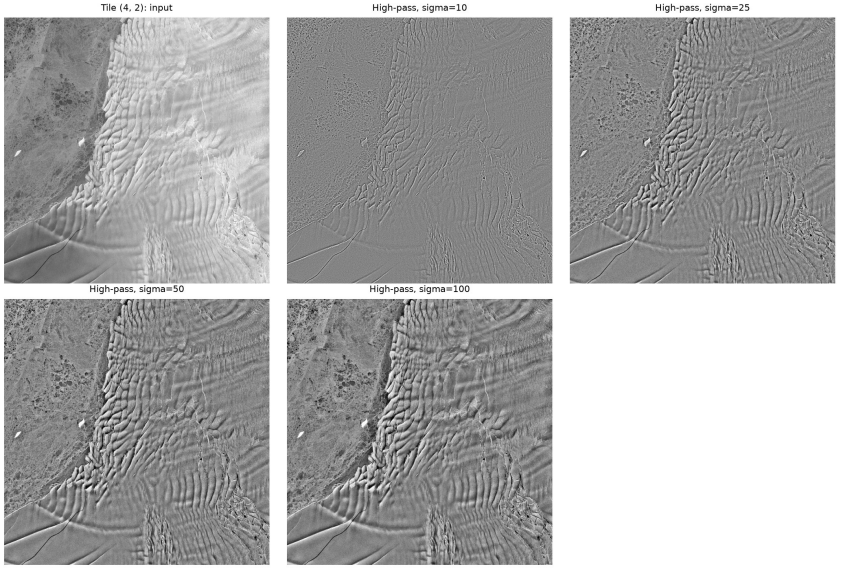
6 Apply and save

Chosen values on frame 024_077_A_140, tiles (4, 2) and (5, 3).

Each tile: input plus four values

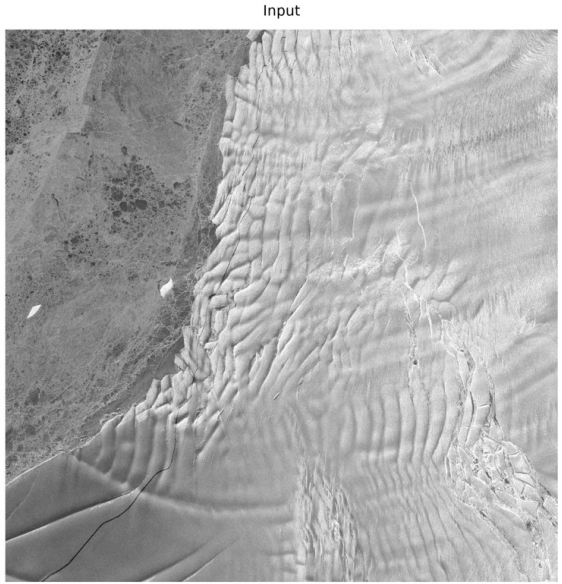


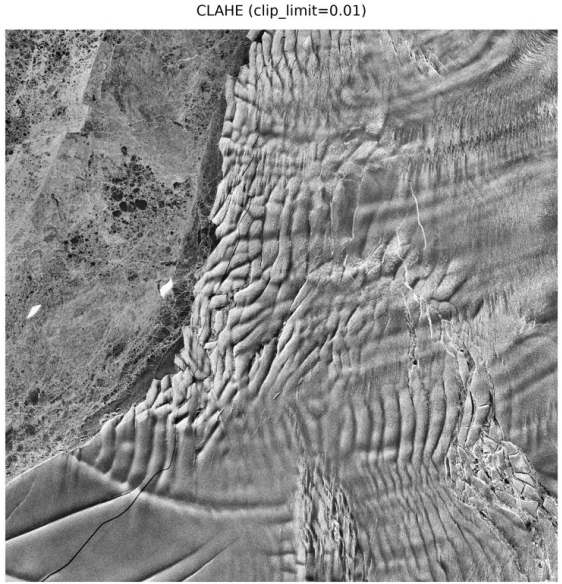
CLAHE, clip_limit 0.005 to 0.05

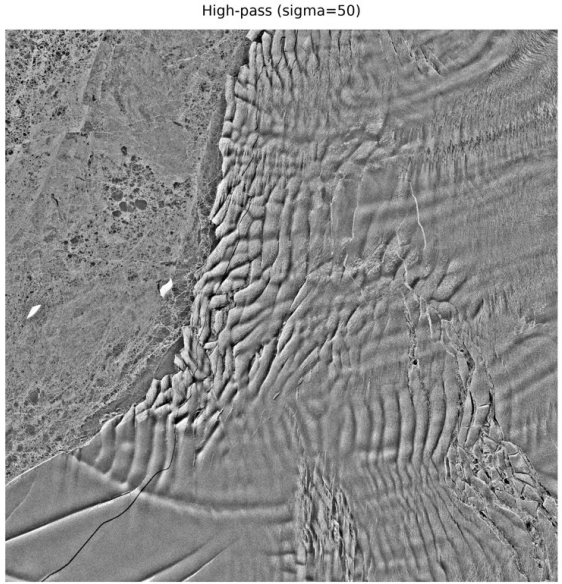


High-pass, sigma 10 to 100

Results







Saved in contrast_enhancement/

clahe/ and high_pass/: processed tile, full size

comparison/: input | CLAHE | high-pass

Next

- Pick clip_limit and sigma by eye
- Then measure with the crevasse labels